12 Here are the first four terms of a quadratic sequence.

3 8 15 24

(a) Find an expression, in terms of n, for the nth term of this sequence.

.....

(3)

The nth term of a different sequence is $2^n + 5$

(b) Show that 36 is **not** a term of this sequence.

.....

.....

(1)

(Total for Question 12 is 4 marks)